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EDUCATION

- **Harvard University** Boston, MA
M.S. in Health Data Science *August 2025 – March 2027*
 - Program requirements complete December 2026 (commencement March 2027); available for full-time start January 2027
 - Coursework: Deep Learning, AI in Medicine, Healthcare Machine Learning, Data Science
- **Emory University, College of Arts and Sciences** Atlanta, GA
B.S. in Applied Mathematics and Statistics; Minor in Computer Informatics; GPA: 3.925 *August 2021 – May 2025*
 - Coursework: Probability and Statistics, Machine Learning, NLP, Data Structures, Database Systems

TECHNICAL SKILLS

- **Languages:** Python
- **ML:** PyTorch, LoRA / PEFT, BLIP-2, DenseNet-121, scanpy, AnnData, Hugging Face
- **Data:** held-out evaluation, Kaggle leaderboard, prompt ablations
- **Certifications:** AWS Certified Cloud Practitioner, AWS Certified AI Practitioner, AWS Machine Learning Associate

MACHINE LEARNING PROJECTS

- **speciesOT: Cross-Species Single-Cell Translation** Boston, MA
Graduate Researcher — Mooney Lab & Alvarez-Melis Lab, Wyss Institute *February 2026 – Present*
 - Compared an ICNN optimal-transport map (IMPACT_CellOT) with an scGen latent mean-shift inside one shared autoencoder on a mouse-to-human single-cell task.
 - On a paired held-out-species evaluation, scGen R^2 was **0.85–0.90** with the source species in training versus **0.65–0.67** when that species was held out.
 - Gene-space ranking reversed with the split. 6619 orthologs won R^2 across genes on the in-train split. 1000 HVGs won the reported metrics when that species was held out.
- **TextVQA: Zero-Shot, Prompts, and LoRA Fine-Tuning** Boston, MA
Sole author — Harvard AI in Medicine *May 2026*
 - Fine-tuned BLIP-2 with LoRA on TextVQA. Zero-shot accuracy was **0.078**. An InstructBLIP OCR-augmented prompt reached **0.106**. LoRA $r=8$ OCR-aware fine-tune reached **0.213**.
 - Ablated LoRA rank over r in $\{4, 8, 16\}$. The adapter is a parameter-efficient fine-tune on one A100.
- **Pneumonia Detection from Chest Radiographs** Boston, MA
Sole author — Harvard BST-261 Kaggle competition *May 2026*
 - Reached a Kaggle public score of **0.959** after diagnosing a 14-point generalization gap (99%+ validation versus 85% leaderboard) on about 4,700 training images.
 - Domain-aware augmentation, with no vertical flips and rotation capped at 15 degrees, lifted the leaderboard from 85% to about 96%.
 - Freezing early layers versus full fine-tuning moved the leaderboard from 85% to 95%+ and mattered more than architecture choice.